Gowin Device Internal User Flash JTAG Programming Guide(Only for GW1N-1 and GW1N-1S)

Overview

The GW1N-1 and GW1N-1S devices support User Flash with 12 Kbytes (48 page x 256 bytes). The features are as follows:

- 100,000 write cycles
- Minimum10 years data retention(+85℃)
- Optional 8/16/32 bits data-in and data-out
- Page size: 256 Bytes
- 3 μA bypass current
- Page write time: 8.2 ms

User Flash Port Signals and Timing

Figure 1 User Flash Diagram

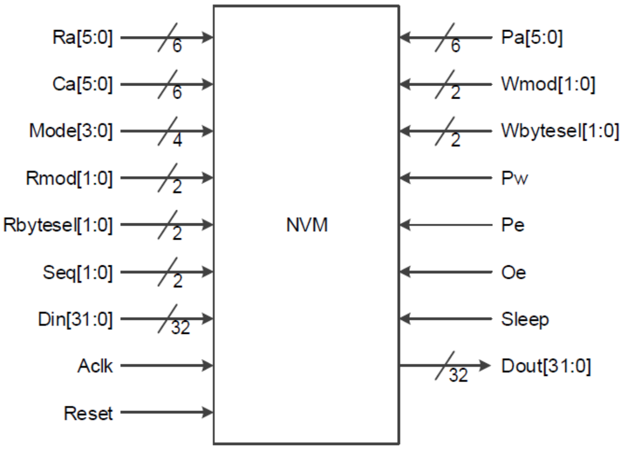


Table 2 User Flash Port Signals

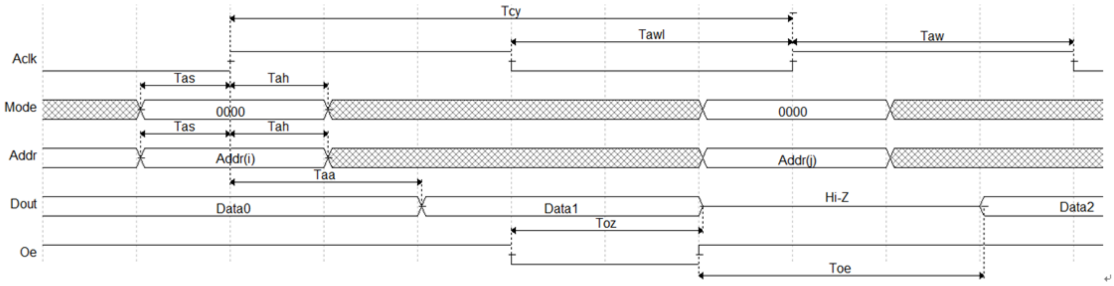
Pin name[1]	I/O	Description
Ra[5:0]	I	X address bus, used to select one row within memory block.
Ca[5:0]	I	Y address bus, used to select one column within memory block.
Pa[5:0][2]	I	I
Mode[3:0]	I	Select operation mode.
Seq[1:0]	I	Control operation sequence.
Aclk	I	Synchronize clock for read-write operations.
Rmod[1:0]	I	Read data bit width selection.
Wmod[1:0]	I	Write data bit width selection.
Rbytesel[1:0]	I	Read data byte selection.
Wbytesel[1:0]	I	Write data bit width selection.
Pw	I	Write Page latch clock.
Reset[3]	I	Reset signal, active-high.
Pe	I	Charge pump enabled.
Oe	I	Data output enable.
Sleep[4]	I	Sleep mode, active-high.
Din[31:0]	I	Data input bus.
Dout[31:0]	O	Data output bus.

Note!

- [1] Port names of Control, address,and data signals.
- [2] Pa signal has the same function as Ca signal, except that Pa signal is used for programming operation of page latch data, and Ca signal is used for other operations related to column selection in Flash.

- [3] The high-level effective time of reset signal is not less than 20ns. Wait for 6μs after that the reset signal changes to low-level, and then move on.
- [4] Save power through flash memory resources entering into sleep mode. Wait for 6μs after that the sleep signal changes to low-level, and then move on.

Figure 3 User Flash Read Timing



Note!

Read operation cycle Seq=0, Addr signal contains Ra, Ca, Rmod, and Rbytesel.

User Flash Programming Method Based on JTAG

1. Create a .fi file: The file extension is .fi, and the example of the file content is as follows:

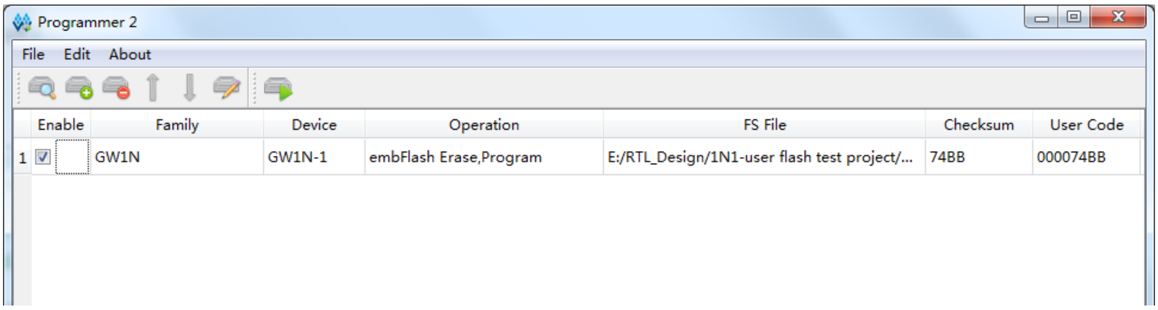
//Copyright (C) 2014-2019 Gowin Semiconductor Corporation.
//All rights reserved.
//File Title: UserFlash Initialization file
//GOWIN Version: V1.9.1Beta
//Part Number: GW1N-LV1QN32C6/I5
//Device-package: GW1N-1-QFN32
//UserFlash: FLASH96K
//Format: Hex
//Created Time: Mon Jun 10 10:39:52 2019

[0:1] ccc01234
[0:2] caa0bbb4
[1:0] 11111111
[1:1] 10101010
[1:2] 01010101
[1:3] 00001111
[3:4] 11110000

Note!

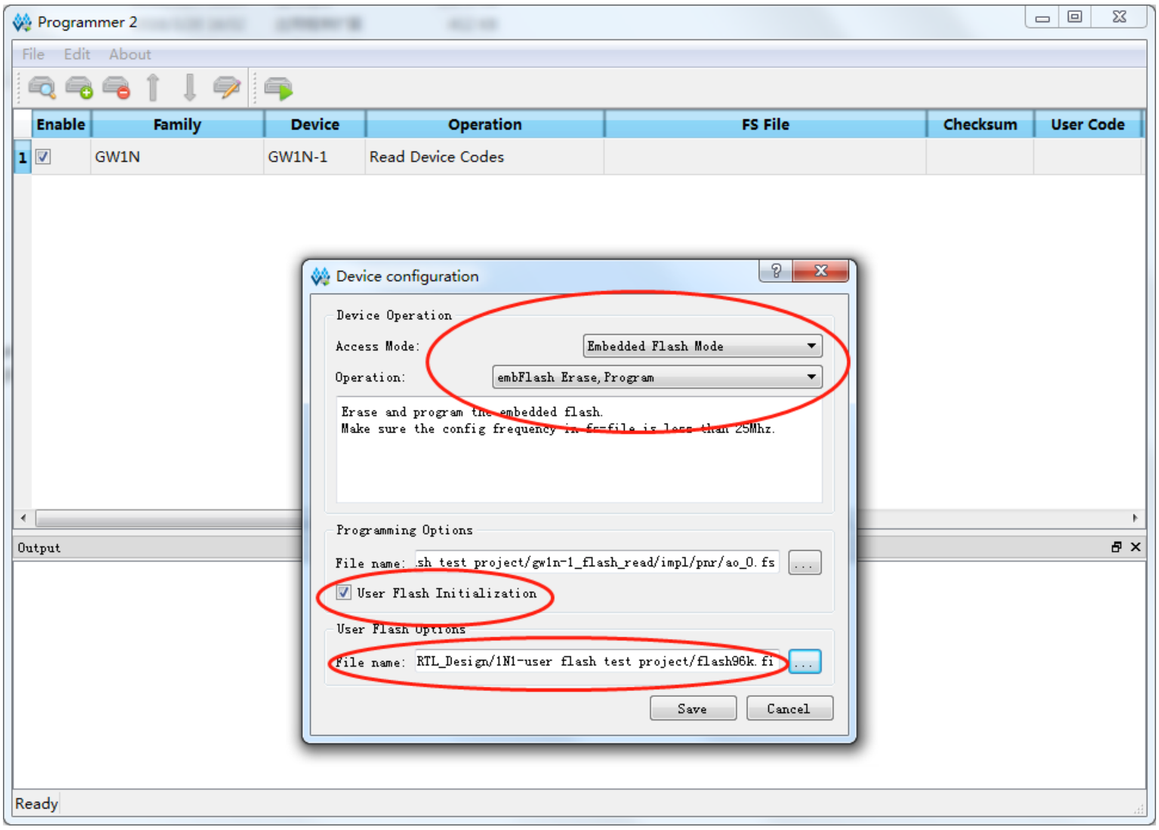
- The first number in the array is Ra address with the bit width of 6 and the address range of 0~47; the second number is Ca address with the bit width of 6 and the address range of 0~63.
- The array is followed by 8 data, hexadecimal and 32 bits wide.

2. Program the .fi file:
   a). Open Programmer;

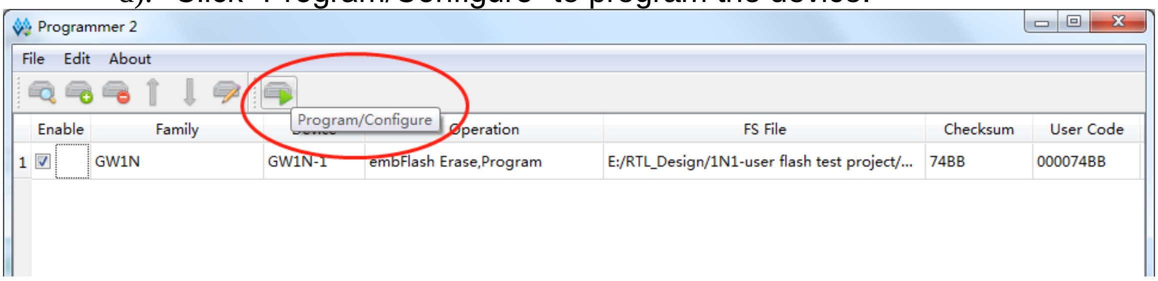


b). As shown in the figure below, select "Embedded Flash Mode" for the Acces Mode, select “embFlash Erase，Program”for the "Operation";

c). Specify a path to the .fs file；Click "User Flash Initialization" with the specific .fi file，and click "Save".



d). Click “Program/Configure” to program the device.



Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly by the following ways.

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Revision History

Date	Version	Description
7/3/2019	1.0E	Initial version published.

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